

MARS ROVER MANIPAL RESEARCH AI TASKPHASE

Task 1: Report

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1. What is research?

Research is a systematic and organized process of investigation to discover, interpret, or revise facts. It's about asking a precise question and following a structured plan to find a credible answer. The ultimate goal is to produce new knowledge, solve a problem, or develop a deeper understanding of a topic. To be considered research, the process must be evidence-based and as objective as possible.

2. Types of research papers

Most research papers fall into one of these common categories:

- **Original research articles:** These are the most common type of research papers. They present new, original findings from the author's own study. They typically follow a standard structure like IMRaD (Introduction, Methods, Results, and Discussion).
- **Review articles:** These provide a critical synthesis of existing research on a topic. They don't present new data but instead summarize and analyze the work of others to identify trends, gaps, and future directions.
- **Case studies:** These are in-depth analyses of a single individual, group, event, or situation. They are common in fields like medicine, psychology, and social sciences.
- **Methodology papers:** These papers present a new experimental or analytical method, test, or procedure. The focus is on the new method itself, not necessarily the findings it produces.
- **Short communications/letters:** These are brief reports of significant, time-sensitive findings. They are shorter than full articles and are intended for rapid publication.

3. The publication process

Researchers typically publish their work in academic journals or conference proceedings. The process is rigorous and designed to ensure quality.

Where papers are published:

- **Academic journals:** These are periodical publications where scholars publish their work. They can be broad (e.g., Nature, Science) or highly specialized (e.g., Journal of Machine Learning Research).

- **Conference proceedings:** In fields like Computer Science, conferences are a primary venue for publishing cutting-edge research. Papers are submitted, peer-reviewed, and if accepted, presented at the conference and published in the official proceedings.

The general process:

1. **Manuscript preparation:** The researcher writes the paper according to the target journal or conference's guidelines.
2. **Submission:** The manuscript is submitted to the publication through an online portal.
3. **Peer review:** The editor sends the manuscript to several independent experts in the same field (peers). They review the paper for validity, significance, and originality. This can be single-blind (reviewers know the author's identity) or double-blind (neither party knows the other's identity).
4. **Decision and revision:** Based on the reviews, the editor makes a decision: Accept, Accept with Revisions, Revise and Resubmit, or Reject. Most often, authors are asked to make revisions.
5. **Publication:** Once accepted, the paper goes through copyediting and typesetting before being published online or in print.

4. How to evaluate a publication venue

Choosing where to publish is critical. The quality of a journal or conference is evaluated using several parameters:

- **Impact Factor (IF) & CiteScore:** These metrics measure the average number of citations received by articles in a journal over a specific period. A higher number generally indicates greater influence.
- **Acceptance rate:** This is the percentage of submitted papers that are accepted for publication. Highly prestigious conferences and journals often have very low acceptance rates (e.g., <20%), indicating high selectivity.
- **Indexing:** Reputable venues are indexed in major academic databases like Scopus, Web of Science, PubMed, and Google Scholar. This ensures the work is visible and accessible to other researchers.
- **Peer review process:** A rigorous, double-blind peer review process is considered the gold standard for ensuring quality and impartiality.
- **Editorial Board:** The reputation of the scholars on the editorial board is a strong indicator of the publication's standing in the academic community.

5. Top conferences in AI and ECS

In Computer Science, conferences are often considered as prestigious as top journals. The CORE ranking system grades conferences from C (*lowest*) to A* (*highest*). Here are some of the most respected A* conferences for AI and ECS.

Artificial Intelligence (AI):

- ***NeurIPS:*** Conference on Neural Information Processing Systems
- ***ICML:*** International Conference on Machine Learning
- ***ICLR:*** International Conference on Learning Representations
- ***AAAI:*** AAAI Conference on Artificial Intelligence
- ***CVPR:*** Conference on Computer Vision and Pattern Recognition

Electronics & Computer Science (ECS):

- ***ISCA:*** International Symposium on Computer Architecture (Computer Architecture)
- ***ASPLOS:*** International Conference on Architectural Support for Programming Languages and Operating Systems (Interdisciplinary)
- ***OSDI/SOSP:*** Symposium on Operating Systems Design and Implementation / ACM Symposium on Operating Systems Principles (Operating Systems)
- ***SIGCOMM:*** ACM Special Interest Group on Data Communication (Computer Networks)